

Report on Social Media Data Sources for GESICA Project

Introduction

The Interreg project called *Gestion de Situations de Crises Sanitaires* (GESICA) aims to develop innovative tools for the effective management of health crises across a region extending from the Canton of Geneva to the Bourgogne-Franche-Comté region. Within this initiative, social media data could be leveraged to identify weak signals and develop machine learning models to predict pandemic dynamics.

This report provides an overview of potential social media platforms from which data may be extracted. For each platform, we discuss studies and data extraction methods relevant to the GESICA project. Specifically, we aimed to identify potential methods and pipelines that could be used to retrieve historical data from the COVID-19 period and related to the region of interest.

Social Media Data Sources

X (Twitter)

X was arguably the most widely used social media platform for research during the COVID-19 pandemic. Notably, we identified a study whose approach closely aligns with the objectives of our project [1]. The authors of this article leveraged tweets from X to predict daily COVID-19 case counts in Australia. Data was extracted using the Twitter API, primarily relying on geotagged tweets. Other relevant studies introduced datasets consisting of tweets associated with COVID-19-related keywords [2, 3, 4]. However, in accordance with Twitter's Terms and Conditions, only Tweet IDs are shared. The full text of the tweets must therefore be retrieved through a process known as *tweet hydration*, using the X API.

Exploiting these datasets to retrieve tweets during the COVID-19 pandemic and related to the region of interest, requires to rehydrate a large volume of tweet IDs, filter them to retain only those with geolocation metadata, and then restrict the dataset to tweets originating from that area. The datasets described above typically cover the entire world and include hundreds of millions of tweet IDs. Among them, a relevant dataset we identified is [2], which comprises 21 million tweets (415,434 of which are in French) and was specifically curated to ensure the presence of geolocation information. Consequently, rehydrating at least several million tweets will likely be required to obtain a sufficiently large and relevant subset for analysis.

A more effective approach may involve working with datasets that include metadata such as tweet language and country, as for instance [4]. Filtering Tweet IDs based on these attributes could help reduce the number of unnecessary API requests. However, a closer examination

of such datasets is necessary to ensure they contain a sufficient number of Tweet IDs corresponding to the region and time period of interest.

Historically, X offered generous API rate limits, which contributed to its widespread use in academic research. However, recent policy changes have significantly restricted access to its data. Although it is difficult to precisely estimate the number of tweets that can be rehydrated per month under the various access, current pricing suggests that the Basic access (USD 200/month) allows up to 15,000 tweet lookups per month, while the Pro access (USD 5,000/month) permits up to one million tweet lookups per month [5].

Finally, it is possible to apply for academic research access to public X data [6], which could provide more flexible or cost-effective access under certain conditions

META

Social media platforms owned by Meta, (i.e., Facebook and Instagram) have been significantly less utilized than X by researchers during the COVID-19 pandemic compared to Twitter. Existing studies that leverage Facebook or Instagram data in the context of COVID-19 primarily focused on sentiment analysis or the spread of misinformation. These articles typically relied on a tool known as the CrowdTangle API, which enabled the extraction of public content from these platforms.

However, CrowdTangle was officially discontinued in Summer 2024 [7]. Meta has since introduced the Meta Content Library (MCL) as a replacement. Access to MCL is free but restricted to academic and non-profit researchers and requires submitting a formal application for approval. However, even if a viable solution is developed using such resources, translating it into a real-world industrial deployment might be challenging or at least costly, as these programs primarily offer cost-free access for research purposes only.

Although relatively small, a dataset of Instagram posts related to the COVID pandemic has been introduced [8]. However, in agreement with Instagram's Terms & Conditions, only the post IDs were released and rebuilding the dataset of posts might be challenging.

Reddit

Reddit was also a viable source of patient information. COVID-19 cases and patient-reported symptoms were identified from Reddit using classification models. The trend of positive and symptomatic cases identified is highly correlated with the officially reported cases in the US CDC [9]. Mining long-term symptoms [10] and public sentiment [11] from Reddit were also feasible.

The Reddit Data API provides both free and paid access, with the free tier intended for non-commercial purposes such as academic research [12]. In terms of rate limiting, users with free access are allowed up to 100 queries per minute per OAuth client ID, averaged over a 10-minute window to allow for occasional bursts. The pricing model charges third-party app developers \$0.24 for every 1,000 API calls, effective July 1, 2023 [13].

Google Trends

Google Trends data [14] covered web-based searches in monitoring COVID-19. Studies in the US have shown statistically significant correlations between Google Trends and COVID-19 data, and the models exhibit strong COVID-19 predictability [15]. Face masks, Lysol, and COVID-19 stimulus checks had the strongest correlations [16]. Studies in other countries and regions have shown similar results. There is a strong correlation between the frequency of searches for smell-related information and the onset of COVID-19 infection in Italy, Spain, the UK, the USA, Germany, France, Iran, and the Netherlands [17]. High to moderate correlations between Google relative search volumes (RSVs) and COVID-19 cases were found in Taiwan [18] and India [19].

Google Trends could identify optimal timing and geographic focus for implementing effective risk communication strategies among affected populations. Pytrends [20], an unofficial Python API for accessing Google Trends data, is one approach to automate the retrieval of trend reports, but it may face rate limits or provide incomplete data. A more reliable solution is to automatically use the download button in Google Trends and download the CSV files.

Telegram

To the best of our knowledge, only a limited number of studies related to COVID-19 and exploiting data coming from Telegram have been conducted. Most of them are related to topics such as misinformation spreading. However, it is worth noting that a multimodal dataset containing data related to posts on public channels has been presented [21]. Studies that extract publicly available data from Telegram typically rely on an API called *Telethon*. This API allows access to the message history of public channels, as well as public groups, provided the user joins them.

Therefore, by identifying Telegram channels and public groups with a clear focus on residents of the region of interest (e.g., local sports clubs), it may be technically feasible to extract historical data relevant to that area. Nevertheless, this approach is not without risks. Notably, deleted messages cannot be recovered. Given the retrospective nature of the analysis, the proportion of deleted content may be substantial, potentially limiting the potential relevance of the retrieved data.

Telethon is free to use, but its rate limits are, to the best of our knowledge, not fully documented in official sources.

Global Public Health Intelligence Network

During our research, we also came across an early warning system for potential public health threats worldwide, provided by the Public Health Agency of Canada [22]. Although this tool is free and might be really relevant for our project, some sources mentioned that this system has been shut down [23].

Conclusion and next steps

As discussed in this report, highlighted in a recent study [24] and illustrated in Fig. 1, access to social media APIs was significantly restricted between 2021 and 2023. These limitations have likely been driven by a desire to regain control over data usage, particularly in the wake of the Cambridge Analytica scandal, and by an increasing awareness of the economic value of user-generated content, especially for training large language models.

In response to the growing restrictions on data access for researchers, the European Union enacted the Digital Services Act (DSA), which mandates that Very Large Online Platforms and Very Large Online Search Engines grant access to both public and private platform data for qualified researchers investigating systemic risks. Consequently, in compliance with the DSA, several platforms have launched dedicated research access programs.

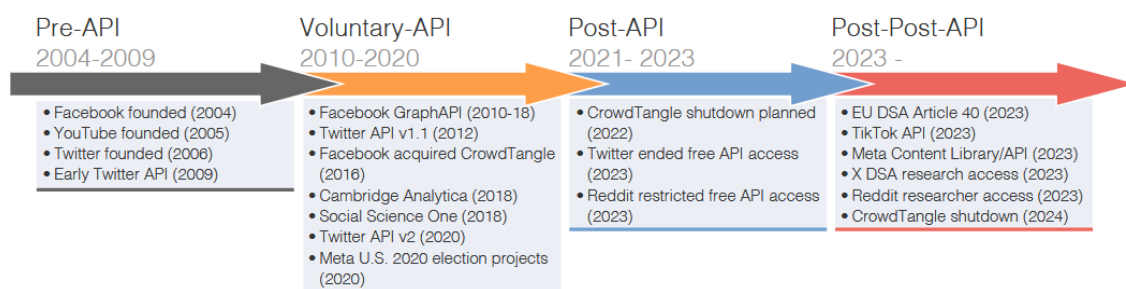


Fig. 1 Different eras of data access to social media platforms. Source: [24].

However, as thoroughly discussed in [24], these programs face several limitations, including excessive administrative burden, opaque and inconsistent application procedures, and significant delays in communication. Some researchers have reported waiting weeks or even months for a response, if any is received at all.

Although accessing data from social media presents challenges, several strategies exist to overcome these limitations. While some of these approaches may be time-consuming, many are cost-free. Despite the aforementioned problems, it seems relevant to apply for research access to public X data and as well to MCL. Due to its user-generated content and accessible data, Reddit is a valuable supplementary source for monitoring self-reported symptoms. While direct access to historical posts is constrained by Reddit's API limitations, recent discussions within geographically relevant subreddits can still support localized analyses and monitoring efforts related to public health topics [25]. Additionally, data from Google Trends can be extracted freely and may provide useful complementary signals. Furthermore, while the outcome of this approach is uncertain, it should be interesting to try to exploit Telegram's channels and public groups, through the use of Telethon API.

Beyond these cost-free options, it may be feasible to allocate financial resources to facilitate data extraction. First, it may be possible to leverage Tweet ID datasets that include additional metadata such as language or country. If these datasets contain a sufficient number of tweets related to the region of interest, basic access to the X API may be adequate to hydrate enough content for constructing a meaningful dataset.

Finally, although potentially costly, commercial tools such as Meltwater could be employed to streamline the data collection process and reduce the manual workload.

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